

The bicaption package*

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Abstract

This package supports the typesetting of bilingual captions.

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1 The user interface

1.1 Loading the package

`\usepackage` This package will be loaded by

```
\usepackage[\langle options \rangle]{bication} .
```

The options for the `bication` package are the same ones as for the `caption` package and specify settings which are used for the second language *additionally*. In fact

```
\usepackage[\langle options \rangle]{bication}
```

is identical to

```
\usepackage{bication}  
\captionsetup[bi-second]{\langle options \rangle} .
```

1.2 Setting options

`\captionsetup`

```
\captionsetup[bi]{\langle options \rangle}
```

do setup options which will be used for bilanguage captions *additionally* to the ones which are setup for the specific floating environment.

```
\captionsetup[bi-first]{\langle options \rangle}
```

do setup options which will be used for the *first* heading of the bilanguage captions *additionally* to the ones which are setup for the specific floating environment and the ones which are setup by `\captionsetup[bi]{...}`.

```
\captionsetup[bi-second]{\langle options \rangle}
```

do setup options which will be used for the *second* heading of the bilanguage captions *additionally* to the ones which are setup for the specific floating environment and the ones which are setup by `\captionsetup[bi]{...}`.

Options specified with `\usepackage[...]{bication}` and `\captionsetup[bi...]{...}` will override the ones specified by `\captionsetup{...}` and `\captionsetup[figure]{...}` (same for 'table'). So finally we have the following order how settings for bilingual captions are applied:

1. Global settings (`\usepackage[...]{caption}` and `\captionsetup{...}`)
2. Environmental settings (`\captionsetup[figure -or- table]{...}`)
3. Local settings (`\captionsetup{...}` inside figure or table environment)
4. Custom 'bi' settings (`\captionsetup[bi]{...}`)

5. Custom ‘bi-first’ resp. ‘bi-second’ settings (`\usepackage[...]{bication}` and `\captionsetup[bi-first]{...}` resp. `\captionsetup[bi-second]{...}`)

An example:

```
\usepackage[labelsep=quad,indentation=10pt]{caption}
\usepackage[labelfont=bf]{bication}
\captionsetup[table]{labelfont=it,position=top}
```

causes the second heading of the bilingual caption inside `table` environments to be typeset with the settings

```
labelsep=quad,indentation=10pt,position=top,labelfont=bf.
```

1.3 Additional options

These options are available additional to the ones offered by the `caption` package:

<code>language=</code>	<code>language=</code>	Sets the language of the caption, e.g. <code>\usepackage[english]{bication}</code> will typeset the second caption of bilingual captions in English. (The language will be set with <code>\selectlanguage</code> internally, so the <code>babel</code> package must be loaded for using this option.)
<code>bi-lang=</code>	<code>bi-lang=</code>	Causes a selection of the headings of bilingual captions. <code>\captionsetup{bi-lang=both}</code> will cause that both caption headings are being typeset. (This is the default.) <code>\captionsetup{bi-lang=first}</code> will cause that only the <i>first</i> heading is being typeset, and <code>\captionsetup{bi-lang=second}</code> will cause that only the <i>second</i> heading is being typeset.
<code>bi-singlelinecheck=</code>	<code>bi-slc=</code>	Switches the common single-line-check on or off, i.e. when switched on only a single check will be done for both captions, and the result will affect both captions afterwards. So if only one caption is longer than a single line, both captions will be treated as if they are longer than a single line, even if the second one isn't. (The default is <code>on</code> .)
<code>bi-swap=</code>	<code>bi-swap=</code>	<code>\captionsetup{bi-swap}</code> will swap the primary and secondary language, making the first language the second one and vice versa. (The default is <code>false</code> .)

1.4 The `\bicaption` commands

`\bicaption` Bilingual captions will be typeset by

```
\bicaption[⟨list entry #1⟩]{⟨heading #1⟩}  
          [⟨list entry #2⟩]{⟨heading #2⟩}  
\bicaption*{⟨heading #1⟩}{⟨heading #2⟩}
```

The `\label` should be placed either after this command, or inside the first heading.

`\bicaptionbox` Bilingual caption boxes will be typeset by

```
\bicaptionbox[⟨list entry #1⟩]{⟨heading #1⟩}  
              [⟨list entry #2⟩]{⟨heading #2⟩}  
              [⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}  
\bicaptionbox*{⟨heading #1⟩}{⟨heading #2⟩}  
              [⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}
```

The `\label` should be placed inside the first heading.

(For a description of the optional parameters `⟨width⟩` and `⟨inner-pos⟩` please take a look at the caption package documentation, `\captionbox`.)

If the `subcaption` package is loaded, these commands are available additionally:

`\bisubcaption` Bilingual sub-captions will be typeset by

```
\bisubcaption[⟨list entry #1⟩]{⟨heading #1⟩}  
              [⟨list entry #2⟩]{⟨heading #2⟩}  
\bisubcaption*{⟨heading #1⟩}{⟨heading #2⟩}
```

The `\label` should be placed either after this command, or inside the first heading.

`\bisubcaptionbox` Bilingual sub-caption boxes will be typeset by

```
\bisubcaptionbox[⟨list entry #1⟩]{⟨heading #1⟩}  
                  [⟨list entry #2⟩]{⟨heading #2⟩}  
                  [⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}  
\bisubcaptionbox*{⟨heading #1⟩}{⟨heading #2⟩}  
                  [⟨width⟩][⟨inner-pos⟩]{⟨contents⟩}
```

The `\label` should be placed inside the first heading.

(For a description of the optional parameters `⟨width⟩` and `⟨inner-pos⟩` please take a look at the subcaption package documentation, `\subcaptionbox`.)

1.5 A sample document

```
\documentclass[english,ngerman]{article}
\usepackage{selinput}
\SelectInputMappings{adieresis={ä},germandbls={ß}}

\usepackage{babel}
\usepackage[lang=english,font=it]{bicaption}
\usepackage[format=hang]{subcaption}

\begin{document}

\begin{figure}[!htb]
  \centering
  \bisubcaptionbox
    {Teilabbildung A\label{fig:test:A}}
    {Subfigure A}[0.4\textwidth]{IMAGE}%
  \quad
  \bisubcaptionbox
    {Teilabbildung langer Titel B\label{fig:test:B}}
    {Subfigure long title B}[0.4\textwidth]{IMAGE}%
  \bicaption{Deutscher Titel}{English Title}
  \label{fig:test}
\end{figure}

\captionsetup{bi-lang=both}

\begin{figure}[!htb]
  \centering
  \bisubcaptionbox[A]
    {Und eine gaaaanz lange Caption: Teilabbildung A}
    {Subfigure A}[0.4\textwidth]{IMAGE}%
  \quad
  \bisubcaptionbox[B]
    {Teilabbildung B}
    {Subfigure B}[0.4\textwidth]{IMAGE}%
  \bicaption[Abbildungsverzeichnistitel]
    {Und eine noch viel viel viel
     längere deutsche Beschriftung: Deutscher Titel}
    {Short English heading}
\end{figure}

\captionsetup{bi-slc=0}

\begin{figure}[!htb]
  \centering
  \bisubcaptionbox[A]
    {Und eine gaaaanz lange Caption: Teilabbildung A}
    {Subfigure A}[0.4\textwidth]{IMAGE}%
  \quad
```

```

\bisubcaptionbox[B]
  {Teilabbildung B}
  {Subfigure B}[0.4\textwidth]{IMAGE}%
\bicaption[Abbildungsverzeichnistitel]
  {Und eine noch viel viel viel
   längere deutsche Beschriftung: Deutscher Titel}
  {Short English heading}
\end{figure}

\captionsetup{slc=0}

\begin{figure}[!htb]
  \centering
  \bisubcaptionbox[A]
    {Und eine gaaaanz lange Caption: Teilabbildung A}
    {Subfigure A}[0.4\textwidth]{IMAGE}%
  \quad
  \bisubcaptionbox[B]
    {Teilabbildung B}
    {Subfigure B}[0.4\textwidth]{IMAGE}%
  \bicaption[Abbildungsverzeichnistitel]
    {Und eine noch viel viel viel
     längere deutsche Beschriftung: Deutscher Titel}
    {Short English heading}
\end{figure}

\end{document}

```

IMAGE
(a) Teilabbildung A
(a) *Subfigure A*

IMAGE
(b) Teilabbildung langer Titel B
(b) *Subfigure long title B*

Abbildung 1: Deutscher Titel
Figure 1: English Title

IMAGE
(a) Und eine gaaaanz lange Caption:
Teilabbildung A
(a) *Subfigure A*

IMAGE
(b) Teilabbildung B
(b) *Subfigure B*

Abbildung 2: Und eine noch viel viel viel längere deutsche Beschriftung: Deutscher Titel
Figure 2: Short English heading

IMAGE
(a) Und eine gaaaanz lange Caption:
Teilabbildung A
(a) *Subfigure A*

IMAGE
(b) Teilabbildung B
(b) *Subfigure B*

Abbildung 3: Und eine noch viel viel viel längere deutsche Beschriftung: Deutscher Titel
Figure 3: Short English heading

IMAGE
(a) Und eine gaaaanz lange Caption:
Teilabbildung A
(a) *Subfigure A*

IMAGE
(b) Teilabbildung B
(b) *Subfigure B*

Abbildung 4: Und eine noch viel viel viel längere deutsche Beschriftung: Deutscher Titel
Figure 4: Short English heading

1.6 Customising lists

`list=` As default both caption texts will be insert into the List of Figures resp. List of Tables. To suppress the second entry just pass the option `list=off` to the `bicaption` package, e.g.:

```
\usepackage[lang=english,...,list=off]{bicaption}
```

`listtype+=` Another option is separating the lists. For that purpose the option

```
listtype+= $\langle list type extension \rangle$ 
```

can be used to tell the `bicaption` package to use a different list for the second caption text. The given value will be appended to the current environment type; for example with `listtype+=X` the list entries will be put into the list responsible for the types `figureX` ($= \text{figure} + X$), `tableX` ($= \text{table} + X$) etc.

Such a $\langle list type \rangle$ can be defined using `\DeclareCaptionType` offered by the `caption` package, but some document classes or other packages offer macros for defining new floating environment types (and their corresponding lists) as well.

A sample document:

```
\documentclass[a4paper]{article}

% Use "ngerman" as 1st language, "english" as 2nd one
\usepackage[english,ngerman]{babel}

% Load the bicaption package with 2nd language set to
% "english", and list type "figureEng" resp. "tableEng"
\usepackage[lang=english,listtype+=Eng]{bicaption}

% Define the new floating environment type "figureEng"
\DeclareCaptionType[fileext=lof2]{figureEng}
    [Figure][List of Figures]
% Define the new floating environment type "tableEng"
\DeclareCaptionType[fileext=lot2]{tableEng}
    [Table][List of Tables]

\begin{document}
\listoffigures      % typeset "Abbildungsverzeichnis"
\listoffigureEnges  % typeset "List of Figures"

\begin{figure}
  \centering
  A placeholder for an image or whatever
  \bicaption{Deutscher Text}{English text}
\end{figure}

\end{document}
```

A different approach is using one list for both languages, but with different formatting. Since the `caption` package does not offer options and commands for customising the format of the lists, one needs an additional package for this purpose, for example the `titletoc` package:

```
\documentclass[a4paper]{article}

% Use "ngerman" as 1st language, "english" as 2nd one
\usepackage[english,ngerman]{babel}

% Load the bicaption package with 2nd language set to
% "english", and list type "figure2" resp. "table2"
\usepackage[lang=english,listtype+=2]{bicaption}

% We load the titletoc package for customizing lists
% Note: Loading titletoc should be done prior defining
% additional floating environments with \DeclareCaptionType
\usepackage{titletoc}

% Define the new floating environment type "figure2"
% Use the same file extension as for "figure" (.lof) here
\DeclareCaptionType[fileext=lof]{figure2}
% Define the new floating environment type "table2"
% Use the same file extension as for "table" (.lot) here
\DeclareCaptionType[fileext=lot]{table2}

% We use the titletoc package for customizing "figure2"
% which is appropriate for the second language captions
\titlecontents{figure2}[3.8em]
{} % no above code
{} % empty numbered entry format
{} % empty numberless entry format
{} % empty filler page format

\begin{document}
\renewcommand\listfigurename
{Abbildungsverzeichnis / List of Figures}
\listoffigures

\begin{figure}
\centering
A placeholder for an image or whatever
\bicaption{Deutscher Text}{English text}
\end{figure}

\end{document}
```